

FE-PIRBN: Feature-enhanced physics-informed radial basis neural networks for solving high-frequency electromagnetic scattering problems

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ABSTRACT

The solution of electromagnetic scattering problems holds significant application prospects in fields such as target detection and electromagnetic metamaterial design. However, efficient prediction of electromagnetic fields under high-frequency and intricate geometric conditions remains one of the most challenging problems. In this study, we proposed a feature-enhanced physics-informed radial basis neural network (FE-PIRBN) for solving high-frequency electromagnetic scattering problems. As a physics-driven neural network model, FE-PIRBN incorporates a PIRBN core network to preserve the local approximating property during the electromagnetic wave propagation process. It combines multi-resolution hash encoding techniques to optimize the capture of high-frequency electromagnetic field features. Under data-free training, the FE-PIRBN enables the accurate prediction of high-frequency electromagnetic scattering at sub-wavelength scales. Based on the testing results of typical cases including the electromagnetic scattering from single and double cylinder, it is evident that FE-PIRBN achieves computational accuracy within a 1.40%-5.82% relative error compared to numerical simulation results and the reference solutions. In increasing frequency scenarios, FE-PIRBN shows excellent generalization capabilities, achieving significant accuracy improvements and more robust convergence in high-frequency electromagnetic scattering tasks, ensuring future reliability and accuracy in wider band applications. Of particular importance is that as the electromagnetic frequency increases, the computational time for numerical methods grows exponentially. In contrast, FE-PIRBN, as a neural network model, maintains a relatively stable parameter size, resulting in significantly faster computation. This presents a promising and potential solution for addressing large-scale electromagnetic problems in the field.

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1. Introduction

High frequency electromagnetic scattering characteristics refer to the amplitude-phase characteristics and their transformation exhibited by a target's scattering field under the influence of high frequency electromagnetic waves. These characteristics vary with changes in physical property parameters such as incidence angle, frequency and polarization [1]. The electromagnetic scattering characteristics of targets serve as the foundation for enhancing radar capabilities in target detecting, tracking and identification [2,3]. Additionally, these characteristics play a crucial role in the development of new camouflage metamaterials [4].

As a typical application scenario of electromagnetic scattering calculations, the core objective of equipment stealth technology lies in minimizing the scattering properties of a target under wave irradiation, specifically its radar cross section (RCS), to reduce the radar echo power in the direction of potential threats. Currently, two primary mechanisms widely employed in stealth technology are scattering and absorption [5]. Scattering involves redirecting radar echoes away from threatening directions. Absorption, on the other hand, involves attenuating the energy of radar waves through the use of electromagnetic wave-absorbing materials. This mechanism effectively reduces both single-station and dual-station radar cross sections (RCS). The wave absorbing capability serves as a key reference parameter for assessing material properties in electromagnetic metamaterial design. By leveraging flexible unit designs and diverse structural compositions, electromagnetic metamaterials can generate various electromagnetic resonances with incident waves at sub-wavelength scales, thereby exhibiting anomalous scattering characteristics [6]. These scattering characteristics can be calculated and predicted to determine whether a material meets specific requirements for a given task. In summary, the accurate prediction of high-frequency target scattering fields plays a vital role in stealth design, radar target detection, and electromagnetic metamaterial design. It serves as a fundamental characteristic in determining the effectiveness of electromagnetic scattering predictions.

Predicting or simulating the wave propagation and electromagnetic scattered fields by numerically solving the Helmholtz equation [7] have become routine [8]. The finite difference method (FDM) [9], finite element method (FEM) [10] and moment method (MOM) [11], are the fundamental numerical methods used in industrial electromagnetic simulation softwares, such as **CST**, **Ansys HFSS**, **feko**, etc. These methods require discretization of the simulation domain and solve the obtained linear systems, which can be computationally intensive. As the incident wave frequency increases and the target scale grows (refer to as high-frequency and large-scale problems), the solution time using these methods can increase exponentially.

To explore new technological approaches, Intelligent Computing [12] has garnered significant attention. It has opened up a new research direction known as "intelligent electromagnetic computing", which is revolutionizing the traditional field of computational electromagnetism that has been developed for over a century. It has been proved that deep learning networks with nonlinear activation functions can approximate any measurable function with arbitrary precision [13,14]. For instance, in the case of antennas, constructing a deep learning network to predict the reflection phase curve based on antenna structure parameters instead of relying on the numerical simulation process can significantly accelerate the subsequent antenna optimization design process by approximately 3~4 orders of magnitude [15].

Data-driven and physics-driven neural networks are two key techniques in the field of intelligent electromagnetic computing. [16,17]. In the data-driven approach, Deep Learning (DL) models function as black boxes that learn the input-output mapping. Deep neural networks (DNNs) exhibit remarkable capabilities in mapping high-dimensional data. For instance, ZHANG and KONG et al. [18,19] have utilized the MOM to derive "intrinsic characteristic parameters" that are independent of the incident wave signature. By learning these parameters, the RCS of Perfect Electric Conductors (PEC) is calculated indirectly. KONG's [19] work introduced an attention mechanism within the neural network, achieving an RCS prediction accuracy of over 98% on the given test set, which is 100 times faster than traditional MOM algorithms. However, data-driven neural network approaches have inherent limitations. The computation required to obtain sufficient training datasets can be prohibitively expensive, and the models may suffer from poor generalization and a lack of conservation to physical laws. Additionally, the interpretability of the data-driven models may be compromised.

Physics-driven neural network models have emerged as a recent focus in the field of electromagnetic computing. These models aim to alleviate the data dependency commonly observed in traditional data-driven approaches by incorporating physical models described by partial differential equations (PDEs), resulting in improved interpretability. One representative algorithm for solving PDEs is the Physics-informed Neural Networks (PINNs) [16], which have demonstrated efficacy in various applications, including fluid dynamics [20,21], geophysical forward modeling [22–24] and inverse problems [25,26]. Inspired by these advancements, physics-driven computational electromagnetism has witnessed rapid development in recent years. For instance, J et al. [27] developed the MaxwellNet to guide optical lens design. They embedded the Helmholtz equation on the U-net framework to establish the mapping between the distribution of dielectric constant and the electromagnetic field. Building upon PINN models, Kazuhiro et al. [28] further analyzed the influence of cavity and surface coating on impedance, proposing an electromagnetic analysis algorithm for multilayer coated cavities. Nguyen-Thanh et al. [29] applied the isoparametric technique to simplify pre-normalize complex computational domains. Ramachandran [30] proposed sparse, physics-based, and partially interpretable neural networks (SPINN) that enhance the interpretability of neural networks for solving PDEs. SPINN can be regarded as a generalized meshless numerical method while retaining the powerful structure of deep learning models. CHEN et al. [31] achieved rapid simulation of electromagnetic characteristics of one-dimensional periodic structures and proposed WaveY-Net based on the U-Net. These advancements highlight the significant progress made in physics-driven computational electromagnetism using neural network models.

However, the research on physics-driven deep learning methods in the field of computational electromagnetism is still in its early stages. While these methods reduce the reliance on extensive data, practical engineering problems involving electromagnetic scattering often pose challenges for physics-driven neural network models. Convergence issues and long training times are observed when dealing with high-frequency physical fields and complex geometries. Recent developments in neural network theory, specif-

ically the neural tangent kernel (NTK) theory, have shed light on the limitations of PINNs in solving partial differential equations with high frequency characteristics or ill-defined computational domains [32,33]. The original NTK theory was proposed by Jacot et al. [34] and demonstrating that infinite width neural networks trained with a sufficiently small learning rate can converge to Gaussian processes [35,36]. This theoretical framework provides insights into the training dynamics of deep learning from a theoretical perspective [37]. Building upon the theory, Wang et al. [38] introduced the NTK theory to PINNs. Inspired by their work, Bai et al. [39] studied the training dynamics of PINNs via the NTK theory. They visualized the evolution of a PINN's NTK throughout the training process and observed that PINNs tend to become local approximators after training. However, when confronted with PDEs exhibiting high-frequency features or ill-posed computational domains, PINNs struggle to make accurate predictions due to difficulties in training them to be local approximators. To address these challenges, a novel neural network structure called the physics-informed radial basis network (PIRBN) was developed, PIRBN naturally possesses specific approximation properties and can converge to a Gaussian process when the neural network width is sufficiently large.

The application of RBN is very wide. In data-driven models, RBN can be used as a classifier. For example, H Rastegar et al. [40] proposed a classifier based on RBFNN and applied Jellyfish search (JS) to select the center of Gaussian function, achieving optimal convergence in EEG classification. F Wurzbeger et al. [41] proposed a deep RBF network initialization scheme based on K-means clustering and covariance estimation, and performed well on image classification and speech emotion recognition tasks. Recently, several studies [42,43] have employed RBF networks as classifiers within deep convolutional neural network architectures, achieving performance comparable to that of the widely used MLP classifiers on benchmark tasks [42]. In the physics-driven model, the initial application of RBN was proposed by Bai [39], demonstrating that PIRBN is more efficient than PINN in solving nonlinear partial differential equations characterized by high-frequency features and poorly conditioned computational domains. The training dynamics of PIRBN were explored using the NTK theory.

Although PIRBN exhibits better capability in capturing high-frequency features when solving PDEs, its learning capacity for complex mappings is limited due to the constraint of a single hidden layer and the fixed centers for all neurons. To address the challenges posed by complex electromagnetic scattering problems, a feasible solution lies in combining embedding methods with the network architecture of PIRBN. Inspired by the success application of embedding methods, particularly encoding input coordinates into neural field representations, Muller et al. [44] implemented the hash coding method and validated its effectiveness various domains such as Gigapixel images, SDF, NRC, and NeRF. The multi-resolution feature provided significant acceleration in the convergence of neural network functional representations, particularly for images represented by NeRF. Therefore, we apply hash encoding to improve the solving precision of PIRBN to provide a more locally aware representation of coordinate values.

Motivated by these successes, to improve the accuracy of forward prediction in scattering field with higher frequency and complex target geometry, we introduce a novel neural network architecture called Feature-Enhanced physics-informed RBF neural network (FE-PIRBN). By incorporating a hash encoding module, we combine the local information from vanilla PIRBN with coordinate position information, thereby enhancing the prediction accuracy of scattering field in scenarios involving higher frequencies and complex targets.

The main contributions of this study are the summarized as the following:

- We propose FE-PIRBN, a novel physics-informed neural network model combines the multi-resolution hash encoding techniques and the radial basis neural network. By integrating the local approximating property of PIRBN and the deep feature learning capability of Hash encoding, FE-PIRBN offers efficient prediction capabilities for high-frequency electromagnetic scattering.
- Addressing the electromagnetic scattering problems governed the Helmholtz equation, a physics-driven training process has been established, encompassing the design of boundary conditions, initial conditions, and the loss optimization. The effectiveness of this approach has been validated in typical scenarios such as multi-cylinder electromagnetic scattering. Test results indicate that FE-PIRBN maintains high accuracy in high-frequency electromagnetic scattering calculations where PINNs fail to converge, and it reduces the computational error by a factor of three to four compared to the original PIRBN in high-frequency problems.
- Through the calculation of electromagnetic wave scattering from an incident frequency range of 95.49 MHz to 716.2 MHz, the applicability and limitations of FE-PIRBN have been investigated. The results demonstrate that FE-PIRBN possesses a high capability for capturing high-frequency signals, and its computational speed is up to $14 \times$ faster than traditional numerical simulation methods. Further analysis reveals that the complex network structure added in this study, compared with existing physics-informed neural networks, significantly improves computational accuracy without causing noticeable performance loss, showcasing the potential of this approach as a novel pathway for electromagnetic scattering calculations.

The remainder of the paper is organized as follows: Section 2 present the methodology, including the overall network architecture of FE-PIRBN and a brief introduction to PIRBN branches and hash coding branches. Section 3 present the experimental results, including the selection of hash coding parameters in electromagnetic cases, the fitting analysis of single and double cylinder electromagnetic scattering fields, and the analysis of time performance. Section 4 presents the conclusion.

2. Methodology

In this section, we present an overview of the architecture of the FE-PIRBN framework. We delve into the specific nuances of two pivotal modules within this framework: firstly, the physics-informed radial basis neural network (PIRBN), which serves as the cornerstone of our approach; and secondly, a meticulous representation of hash encoding, which further enhances the model's capability. Furthermore, we undertake an in-depth analysis of the configuration of hard constrained boundary conditions within the

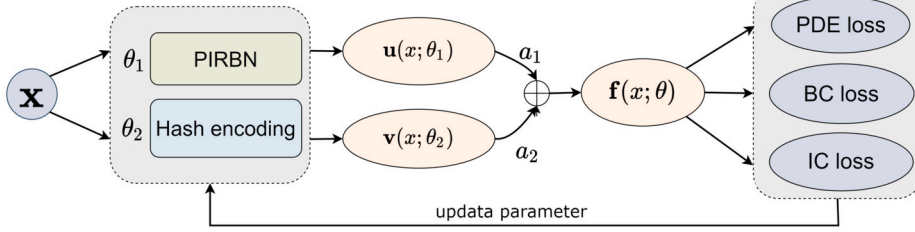


Fig. 1. An overview of the architecture of FE-PIRBN. $\mathbf{X}$ represents the input coordinate array. $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{f}$ represent the PIRBN module, hash encoding module and the weighted network output, respectively. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

perfect matching layer, particularly in light of the limitations posed by the low absorption efficiency inherent in first-order absorption boundary conditions.

2.1. Architecture of the feature-enhanced physics-informed radial basis neural network

Fig. 1 depicts the foundational architecture of FE-PIRBN and outlines the feature transformation process. Within FE-PIRBN, the hash-encoding and PIRBN modules function in parallel, converting the same input coordinates into distinct types of feature information. The hash-encoding module transforms the coordinates into high-dimensional position feature information, while the PIRBN module extracts specific feature information pertinent to the physics of the system. This dual strategy, utilizing the same input coordinates, aims to comprehensively capture coordinate features from both positional and specific perspectives, thereby enhancing the network's ability to model complex physical phenomena.

The hash-encoded location features are trained through a fully connected network and are subsequently incorporated into the PIRBN output, serving as feature reinforcement elements. Upon the determination of the definitive real and imaginary components of the solutions, the network parameters are refined based on the loss term, mathematically formulated in Eq. (3). Notably, the fusion of the two distinct modules is achieved through the implementation of a weight addition strategy, ensuring a seamless integration that enhances the overall performance of the system.

$$\mathbf{f}(\mathbf{x}; \theta) = a \cdot \mathbf{u}(\mathbf{x}; \theta_1) + (1 - a) \cdot \mathbf{v}(\mathbf{x}; \theta_2), \quad (1)$$

As illustrated in Eq. (1), the function $\mathbf{f}(\mathbf{x}; \theta)$ serves as the output of the entire network, representing the ultimate predicted scattering field in the form of a set of complex numbers, specifically $\mathbf{f} = \mathbf{f}_1 + i\mathbf{f}_2$. The input coordinate array is denoted by $\mathbf{x} = x_1, x_2, \dots, x_n$, while the parameter set for the two network modules is given by $\theta = \theta_1, \theta_2$. The variables $\mathbf{u}$ and $\mathbf{v}$ correspond to the feature output of the PIRBN module and the hash encoding module, respectively, and are also expressed as a set of complex numbers with a shape identical to that of $\mathbf{f}$. The parameter a denotes the weight ratio of different modules in the network output, which is set to be 0.8 based on empirical testing.

Here is a brief description of the loss item settings of the network. It begins by highlighting that physics-informed neural networks are prone to local optimality, which hinders their ability to resolve ill-conditioned equations and obtain higher frequency solutions. The continuity inherent in time domain problems exacerbates the challenge of concentrating on the frequency domain. Our primary focus lies in addressing these issues within the frequency domain. To achieve this, the time component is eliminated from the Maxwell equations, necessitating mathematical manipulation via the Fourier transform [45]. The present paper concentrates on resolving the two-dimensional frequency domain's electromagnetic wave equation, known as the Helmholtz equation. It is articulated by the following equation::

$$(\nabla^2 + \varepsilon\sigma\omega^2)\mathbf{E} = \mathbf{f}, \quad (2)$$

Where ∇ denotes the Laplace operator; σ denotes the loss factor, which is set to 1 in this context to simplify the problem, implying a lossless scenario; the dielectric constant is denoted by ε , which is set to 1.0 to represent air; the angular frequency is denoted by ω , from which the frequency of the incident wave is derived; the product $\varepsilon\omega^2$ can be expressed by the wave number k , which corresponds to the number of cycles within a length 2π , with larger k values indicating higher frequencies; $\mathbf{f}$ denotes the excitation of the source, which is set to 0 for external field wave sources.

The physical loss term is manifested in the back-propagation update parameter, as depicted in Fig. 1. It is divided into three parts: PDE loss, BCs loss, and ICs loss, as defined by Eq. (3):

$$Loss = \underbrace{\frac{1}{N_{pde}} \sum_{i=1}^{N_{pde}} ((\nabla^2 + \varepsilon\sigma\omega^2)E(\theta, x_i) - f_i)^2}_{\text{Interior PDE loss}} + \underbrace{\frac{1}{N_{BCs}} \sum_{j=1}^{N_{BCs}} (E(\theta, x_j) - u_b(x_j))^2}_{\text{Supervised loss on boundary}} + \underbrace{\frac{1}{N_{ICs}} \sum_{k=1}^{N_{ICs}} (E(\theta, x_k) - u_{in}(x_k))^2}_{\text{Supervised loss on initial}} \quad (3)$$

where N_{pde} represents the number of collocation points within the domain, N_{BCs} denotes the number on the boundaries, and N_{ICs} signifies the number at the initial state. u_b refer to the boundary values, and u_{in} is the initial values. In defining boundary and initial conditions, it may be necessary to apply operator constraints or field derivatives. Upon obtaining the network output, we employ

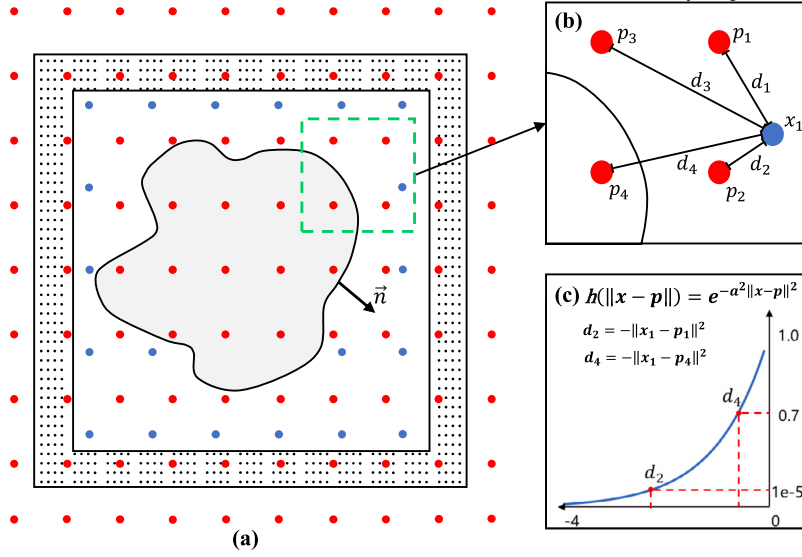


Fig. 2. Relationship between a given neuron and a coordinate point. (a) Shows the distribution of points in the target domain, where the red dot signifies the neuron's center and the blue dot represents the input coordinate. (b) delineates the local relationship. (c) demonstrates the relationship between the network output and the distance $d = -\|x - p\|^2$. In (b), only points x_1 are closer to neuron p_1 and p_2 . Upon activation by the radial basis function, p can more effectively capture the positional features of the points.

Eq. (3) to optimize the aforementioned network parameter θ , thereby achieving a solution that increasingly adheres to the physical law.

2.2. The PIRBN and Hash-encoding module of FE-PIRBN

In this subsection, we delve into the pivotal constituents of the FE-PIRBN framework, specifically the PIRBN module $\mathbf{u}(\mathbf{x}; \theta_1)$ and the Hash-encoding module $\mathbf{v}(\mathbf{x}; \theta_2)$. Additionally We scrutinize the configuration of hard constrained boundary conditions for the perfect matching layer (PML), considering the low absorption efficiency of the first-order absorption boundary conditions.

2.2.1. Physics-informed radial basis network

The radial basis function (RBF) depends solely on the Euclidean distance between the neuron's center $\mathbf{p}$ and the input coordinate $\mathbf{x}$. The function's characteristic is that it imparts to the neural network a favorable specific approximation property prior to training. Given the potential for significant local amplitudes in high-frequency electromagnetic problems, it is imperative for the model to exhibit robust specific feature extraction capabilities. Radial basis networks, initially proposed by Broomhead and Lowe [46], are single-layer neural networks. Originally, The RBF in these networks was originally applied as an activation function [47]. The Gaussian function is the most commonly utilized RBF, expressed as:

$$\mathbf{h}(\mathbf{x}) = \mathbf{h}(\|\mathbf{x} - \mathbf{p}\|) = \exp(-a^2 \|\mathbf{x} - \mathbf{p}\|^2), \quad (4)$$

where shapes the Gaussian function. Other functions such as the inverse quadratic function and spline function can also serve as RBFs. Owing to its simple form, ease of implementation, excellent approximation performance, and smooth characteristics, the Gaussian function is adept at capturing the coordinate relationship between neurons and input points. Fig. 2 illustrates the calculation process of the RBF. As previously mentioned, the RBF output size is determined by the input coordinates and the given center point. Fig. 2(a) depicts the region to be solved, while Fig. 2(b) shows a localized magnification. The red dots represent the distribution of the given centroid $\mathbf{p}$, which corresponds to the distribution of the PIRBN neurons, and the blue dots represent the input coordinate points $\mathbf{x}$. The output of this neural network represents the square of the Euclidean distance between the input coordinates $\mathbf{x}$ and the neuron's center $\mathbf{p}$. Upon activation by a Gaussian function, the closer the distance, the larger the output; conversely, and farther the distance, the smaller the output, i.e., the closer the point in the region to the neuron, the easier it is to be activated. The neurons are uniformly distributed throughout the region and can better capture specific features in all input coordinates. This paper adopts the Gaussian function as RBF. Thus, the mapping between inputs and outputs can be expressed as Eq. (5):

$$\mathbf{u} = \frac{1}{\sqrt{n}} \sum_{i=0}^n \theta_{1_i} h_i(\mathbf{x}), \quad (5)$$

Where n denotes the width of the radial basis network, h_i is the radial basis function defined by Eq. (4) and θ_i is the trainable parameter. We set a to a constant value of $3 \sim 8$ in the scattering problem based on prior testing [39].

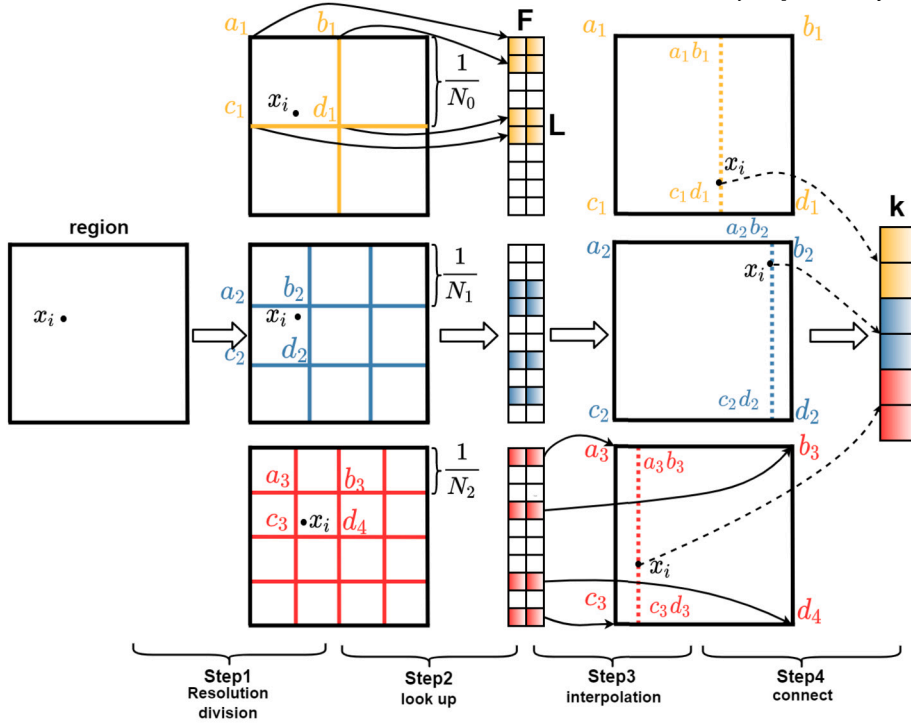


Fig. 3. An overview of 2D multi-resolution hash-encoding mechanisms. Step 1 Division: The region to be solved is divided into different resolution levels, and the diagram is divided into three resolution levels. Step 2 Query: In the current resolution, the vertex of the rectangular grid where the point is to be solved is mapped to a hash feature by the hash function $h(x)$. Step 3 interpolation: the hash features of the vertices are interpolated to obtain the features of the points to be solved. Step 4 connection: the interpolation features of different resolutions are concatenated into the hash feature vector k of the points to be solved.

Consequently, the PIRBN network module is represented as Eq. (6):

$$\mathbf{u}(\mathbf{x}; \theta_1) = \frac{1}{\sqrt{n}} \sum_{i=0}^n \theta_{1_i} \exp(-a^2 \|\mathbf{x}_i - \mathbf{p}_i\|^2), \quad (6)$$

where θ_1 is a trainable parameter, and the final output requires neural network training. The RBF network is characterized by its minimal layer depth; Only one layer of network is necessary to achieve the ideal result. The calculated $\mathbf{h}$ is input into the single-layer network, and the entire network's computational cost is confined to the single-layer network's weight calculations.

2.2.2. Multi-resolution hash encoding to the input coordinates

To improve the approximation accuracy of the neural network for solving partial differential equations (PDEs) with a constant amount of input data, we employ trainable hash encoding. This method expands the input coordinates from two dimensions to a higher dimension, thereby extracting more features. A key advantage of hash encoding is its automatic focus on details, independent of the specific task. In the context of electromagnetic scattering, given that the incident wave is known, hash encoding, in conjunction with the RBF network, provides insights into both the electromagnetic properties and positional attributes of specific coordinate points, creating a synergistic effect. The method involves decomposing the target region into multi-resolution levels, applying trainable encoded features, and employing interpolation to represent different input coordinates with high-dimensional features. Fig. 3 outlines the hash coding mechanism.

Step 1 shows the region containing point $\mathbf{x}$ partitioned into three different resolution regions, with point $\mathbf{x}$ located within a specific rectangular region. N_i represents varying resolution levels in a particular direction. Eq. (7) determines the constant b , and Eq. (8) calculates each resolution level. Step 2 indicates that the hash function maps the four vertices of the rectangular region, where $\mathbf{x}$ is derived from two-dimensional coordinate information, to high-dimensional feature information. Eq. (9) defines the hash function used to generate the hash feature. Each vertex is represented by two features F in the diagram. Step 3 utilizes of linear interpolation to compute the features of the current point $\mathbf{x}$ based on the hash features of vertices from different rectangular regions. Step 4 integrates local features from three types of $\mathbf{x}$ to form the final feature for $\mathbf{x}$. Note that the hash map is represented by a neural network, i.e., $\mathbf{z} = \mathbf{z}(\mathbf{x}; \theta'_2)$, where θ'_2 is the network parameter, indicating its trainability.

The input data is partitioned into R resolution values, ranging from T_{min} to T_{max} . Each level of resolution is determined by:

$$b = \exp\left(\frac{\ln N_{max} - \ln N_{min}}{R - 1}\right), \quad (7)$$

$$N_i = \lfloor N_{min} * b^i \rfloor. \quad (8)$$

b is a constant for every solution. For a particular resolution N_i , $\mathbf{L}$ in Fig. 3 is the length of the query vector at the current resolution, and $\mathbf{F}$ is its dimension. We map the vertices of the rectangle where the target point $\mathbf{x}$ is located to the array of query vectors. It must be considered whether the hash features corresponding to rectangular vertices at different resolutions will create hash conflicts. Since the lookup table function of the neural network is used to implicitly implement the hash representation, the gradient optimization process will automatically avoid hash conflicts. For high-resolution mapping, we use the following hash function [48]:

$$\mathbf{m}(\mathbf{x}) = \left(\bigoplus_{i=1}^d x_i \pi_i \right) \bmod K, \quad (9)$$

where $\oplus$ denotes a bit-by-bit XOR operation and π_i is a large prime number. The hash function XORs pseudo-random permutations in each dimension, aiming to mitigate the effect of high latitude on the hash value [49]. Hash conflicts at high resolution can be further eliminated here. The problem dimensions in this paper are set to 2 dimensions. Therefore π_i ($i = 1, 2$) chooses 1 and 2654435761 respectively. After hash mapping, it is sufficient to fit the interpolation to each point according to the steps shown in Fig. 3.

The generated feature vector $\mathbf{k}$ is trained as an input to a fully connected network. The fully connected layer is represented as:

$$\mathbf{v}(\mathbf{x}; \theta_2) = \mathbf{v}(\mathbf{k}; \theta_2'') = \mathbf{v}([z(x_0; \theta_2'), \dots, z(x_n; \theta_2')]; \theta_2''), \quad (10)$$

where θ_2 is the set of θ_2' and θ_2'' , with θ_2' being the hash map network parameter and θ_2'' being the fully connected layer network parameter. Finally, it is necessary to test the hyper-parameters of the hash encoding framework. We primarily focus on the hash table feature length L and feature dimension F . This content will be elaborated in the experimental section.

2.3. Boundary conditions for electromagnetic scattering cases

For electromagnetic scattering scenarios, boundary settings encompass both absorbing boundary conditions (ABCs) and perfectly matched layers (PMLs) [50]. ABCs are suitable for finite-domain scattering scenarios, aligning with numerical methods such as finite differences. They function by directing waves to extend indefinitely without interacting with boundaries, which is effective in one dimension [51]. However, scattering problems are predominantly two-dimensional (2D) or three-dimensional (3D), where the multitude of propagation directions can complicate ABC absorption. PMLs, on the other hand, are an independent class of absorbing boundary layers. As waves penetrate the PML, they are attenuated through absorption. Even if they reflect, they experience exponentially decay after multiple absorptions within the layer. This indicates that the PMLs utilize an artificial material layer to absorb scattered waves external to the computational domain. As depicted in Fig. 2(a), the black dot area at the boundary shows the PML layer, and the points within this layer are also part of the regional points for the physical quantity solution. After applying PML [52], the PDE Eq. (2) transforms as follows:

$$\frac{1}{1 + i \frac{\sigma_x(x)}{\omega}} \frac{\partial}{\partial x} \left(\frac{1}{1 + i \frac{\sigma_x(x)}{\omega}} \frac{\partial E}{\partial x} \right) + \frac{1}{1 + i \frac{\sigma_y(y)}{\omega}} \frac{\partial}{\partial y} \left(\frac{1}{1 + i \frac{\sigma_y(y)}{\omega}} \frac{\partial E}{\partial y} \right) + k_0^2 E = 0, \quad (11)$$

where

$$\begin{aligned} \sigma_x(x) &= \sigma_0 (x_{left} - x)^2 1_{(-\infty, x_{left}]}(x) + \sigma_0 (x - x_{right})^2 1_{(x_{right}, \infty)}(x), \\ \sigma_y(y) &= \sigma_0 (y_{down} - y)^2 1_{(-\infty, y_{down}]}(y) + \sigma_0 (y - y_{up})^2 1_{(y_{up}, \infty)}(y), \end{aligned}$$

with $\sigma_0 = -\ln 10^{-20} / (4d^3/3) \gg 1$, where d is the depth of the PML layer, set to 1 in this study. For the BCs within the PML layer, we apply zero Dirichlet hard boundary conditions.

3. Results

This section evaluates the efficacy of our approach through a series of three test cases that increment in complexity. For all cases, we employ the FE-PIRBN as the primary neural network architecture, leveraging the **deepXDE** framework [53]. The hash encoding module within our network utilizes the hyperbolic tangent (tanh) as the activation function. Neural network training procedures were conducted on a GTX4090 GPU with 24 GB of memory, while numerical simulations using **feko** were performed on an AMD Ryzen 7 5800H processor. Our aim is to establish the capability of the FE-PIRBN method to accurately capture intricate, high-frequency information.

To neutralize the effects of varying parameter settings within the hash encoding module and the diverse methods of coordinate point selection—two critical preprocessing steps—we begin with a scenario involving an incident wave in the absence of a target. We systematically examine and analyze how different parameter configurations and coordinate point selection strategies affect the model's accuracy and convergence rate. Consistent strategies were applied across all case studies.

In the context of scattering cases, we assess frequency bands, which are indirectly represented by the wavenumber k , using four different methodologies: PINNs, PINNs with Hash-encoding (PINN-Hash), PIRBN, and FE-PIRBN. Uniform epoch thresholds were set to terminate training for each method, with evaluations based on the final accuracy—the absolute error between the solution predicted by the neural network and the analytical solution.

Table 1

Basic parameter setting of hash coding module. Columns 2 and 3 represent the number of iterations where the accuracy first converges to the order of $\exp(-2)$ and $\exp(-3)$, respectively. Column 4 indicates the best accuracy for the current parameter Settings. Column 5 represents the average iteration time per 1000 steps.

Parameter setting		First epoch to e^{-2} metrics		First epoch to e^{-3} metrics		best metrics(e^{-3})		Training time per 1000 epochs
L	F	epoch	metrics	epoch	metrics	epoch	metrics	
10	8	2000	3.37	8000	8.07	8000	8.07	30.39 s
10	4	2000	2.74	5000	7.82	8000	3.61	30.66 s
14	8	2000	2.32	4000	9.97	8000	3.75	32.36 s
14	4	2000	3.04	4000	9.73	8000	4.93	29.57 s
8	4	2000	2.74	5000	7.56	8000	3.54	28.49 s
8	8	2000	2.51	4000	9.25	8000	3.22	31.47 s

3.1. Incident field prediction

In the subsequent scattering problem, we adopt the total field decomposition technique, that is, the incident state is defined on the target (detailed in Section 3.2). Therefore, in the problem of fitting incident waves without targets, we consider to construct the Helmholtz equation problem on two-dimensional rectangular field instead of the incident wave fitting problem without target. This case describes the wave phenomenon of an incident wave propagating into a vacuum region and the equation is almost always applied in electromagnetic scattering problems [54].

The equations are set up as follows:

$$\begin{aligned} -u_{xx} - u_{yy} - k_0^2 u &= f(x, y), \quad \Omega = [0, 1]^2, \\ u(x, y) &= 0, \quad (x, y) \in \partial\Omega, \end{aligned} \quad (12)$$

$$f(x, y) = k_0^2 \sin(k_0 x) \sin(k_0 y),$$

where f is the source function, u is the wavefield, k_0 is the wavenumber. An analytical solution for this equation exists:

$$u(x, y) = \sin(k_0 x) \sin(k_0 y) \quad (13)$$

One can refer to Ihlenburg's book [55] for more details.

The above is an expression of the Helmholtz equation. Based on this example, we discuss the setting of network parameters and initial input Settings.

3.1.1. Parameter setting of hash feature table

The first is the parameter setting of hash feature table. Our goal is to enhance the feature collection of PIRBN by using the high-dimensional position features of hash coding to improve the accuracy of the solution. Neural network automatic differentiation is itself a discontinuous derivative. As shown in step.3 of Fig. 3, the high-dimensional position features are formed by interpolation, which is similar to the above discontinuous automatic differentiation scheme, so the automatic differentiation can be maintained when the network parameters are updated by backpropagation.

The main basic parameters considered here include the length of the hash features L , the total number of features F (the product of the number of resolution divisions and the expression of the features per unit of resolution). This problem (with a large number of waves k) is an insurmountable challenge for PINNs due to its vast range of computational domains and high frequency. To solve similar high-frequency problems, PINN has evolved several times, but the code implementation is more complex. For example, in Dong and Li [56] the entire computational domain is divided into multiple subdomains and different networks are used to fit different regions. The task is simple for PIRBN and FE-PIRBN. We tested a representative sample of six different parameter Settings. In the six groups of tests, the number of iteration steps was 8000, and the learning rate was 0.001. In order to reduce the calculation cost of the fully connected layer, the fully connected layer was set to 2 layers, and the number of neurons in each layer was 32. The overall training coordinate point was 900, and the boundary training coordinate point was 120.

Notice the "metrics" herein in Table 1. As the analytical solution of this case is readily obtainable, the comparison between the analytical solution and the neural network prediction solution can be incorporated into the **deepxde** solution process, namely, the "metrics". It is a **deepxde** built-in test standard, generally with a maximum size of 1, and the smaller, the better. The following partial scattering cases are addressed by the absolute error percentage between the neural network prediction solution and the numerical solution due to the difficulty of obtaining the analytical solution.

Table 1 shows the convergence test with different parameter settings of the hash coding module. The first column of parameter Settings shows component combinations of different feature table lengths and dimensions. Columns 2 and 3 represent the number of iterations where the accuracy first converges to the order of e^{-2} and e^{-3} , respectively. Column 4 indicates the best accuracy for training 8000 steps at the current parameter component Settings. Column 5 represents the average iteration time per 1000 steps. The reason for choosing L is to test whether the hash network here can handle hash conflicts better. Since the number of input points is

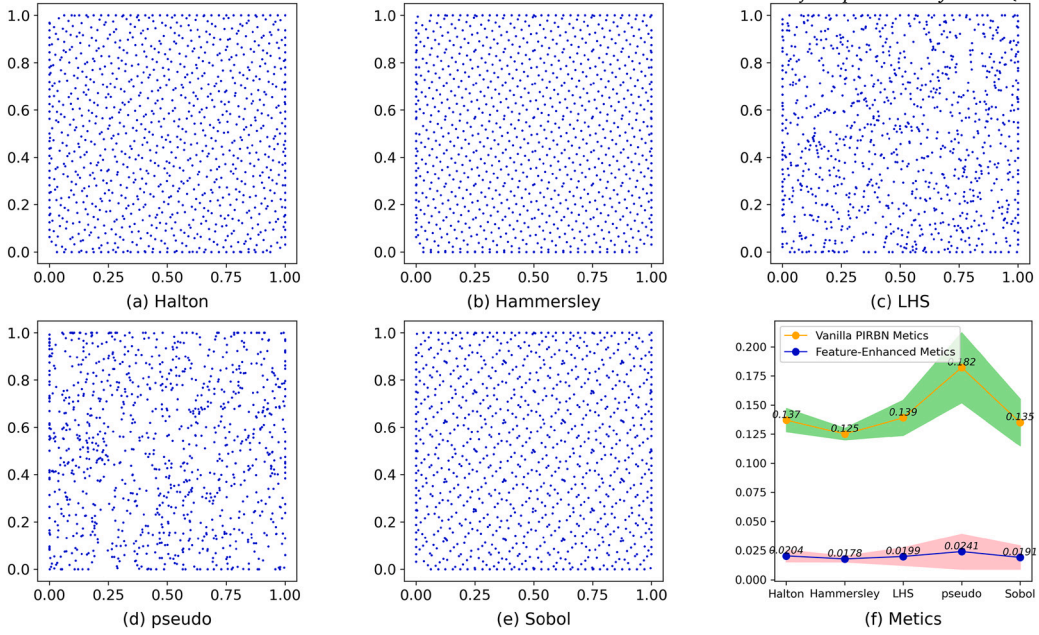


Fig. 4. Schematic representation of five different coordinate point collection methods and their impact on the accuracy of solving the Helmholtz equation.

900, L only needs to be 10 to make the hash query without duplicates for all input points. Therefore, hash collisions occur when L is less than 10. According to the test, the “embedding” query table in the neural network (the hash table is implemented by the query table) can automatically handle the hash conflict, so do not pay much attention to it. Lines 1, 3, and 6 show that the training time per 1000 steps is about the same for different L cases. When $L=8$ and $F=8$, the metrics are improved by 60.10% compared with $L=10$ and $F=8$.

Although, when $L=8$, $F=4$, the accuracy is reduced by 9% compared to $L=8$, $F=8$. However, from the point of view of time cost analysis, every thousand steps, $L=8$, $F=4$ components compute 9% faster. Therefore, based on the above considerations, parameter Settings of $L=8$ and $F=4$ are selected for subsequent cases.

3.1.2. Input distribution settings for initial coordinate points

Different initial point input distribution settings are discussed next. We tested a total of five different distribution methods, including Halton, Hammersley, LHS, pseudo, and Sobol. Fig. 4 demonstrates the point distribution of five different coordinate point collection methods, and it can be seen that the Hammersley method has the most uniform coordinate point distribution. Fig. 4(f) demonstrates the comparison of the solution accuracies of vanilla PIRBN and feature-enhanced PIRBN under different coordinate point collection methods. The green color band indicates the solution accuracy of different point collection methods under vanilla PIRBN, and the pink color band indicates the solution accuracy of different point collection methods under FE-PIRBN. The narrower the color band, the more accurate the solution. The Hammersley method is the most effective in both network frameworks. Therefore only the Hammersley collection strategy is considered in subsequent cases. Fig. 4(f) also shows that in **deepxde**’s “metrics” evaluation criteria, the metrics of FE-PIRBN is improved by an average of about 90% compared to vanilla PIRBN.

Fig. 5 shows the precision comparison between FE-PIRBN and the vanilla PIRBN algorithm when 8400 configuration points were used, hash feature table length $L=8$, dimension $F=4$, and Hammersley point-taking strategy was adopted. (a) denotes the analytical solution of the equation. (b)(d)(e) represent the fitting results in the vanilla PIRBN, PINN, and FE-PIRBN frameworks, respectively. It can be seen that PINN cannot fit the higher wave number case. Because the error of PINN here is too large, the error effect diagram of PINN and analytic solution is not shown. (c)(f) denote the absolute error distributions of the vanilla PIRBN and FE-PIRBN with the analytical solution, respectively. Through the analysis of error effect diagram, the absolute error of FE-PIRBN is 0.07, which is about 84.78% higher than the maximum absolute error of PIRBN 0.46.

3.2. Cylindrical scattering problem of two-dimensional ideal conductor

Electromagnetic optical metamaterials are artificial structures composed of subwavelength building blocks that can achieve unusual optical properties [57]. Current metamaterial designs rely on effective medium theory (EMT) [58–60]. The periodic or aperiodic combinations of dielectric nanocylinders can form a variety of material clusters. The scattering problem of the ideal conductor cylinder becomes the basis of the above metamaterial problems, which can be used as the scattering problem research of partial metamaterials and the inverse metamaterial design. The case allows to solve the ideal conductor scattering problem for 2D, where the radius of the cylinder is R . The problem is first demonstrated by the single-cylinder case which is easy to generate an analytical solution. We also used **feko** software to perform numerical simulations under the same conditions to test approximate differences between the MOM

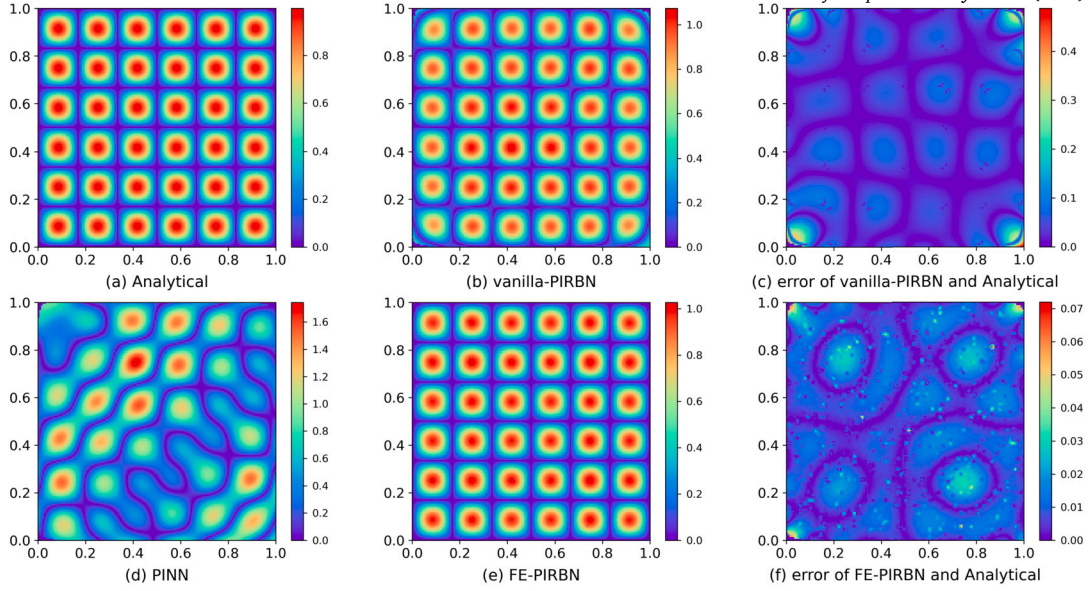
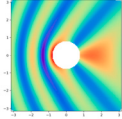
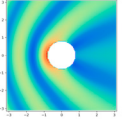
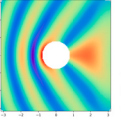
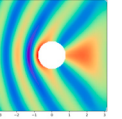
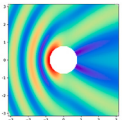
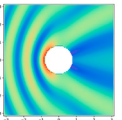
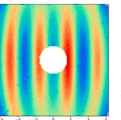
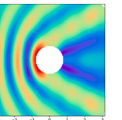
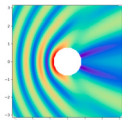
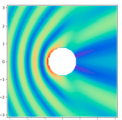
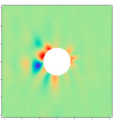
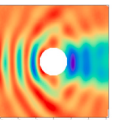


Fig. 5. The non-target incident wave field is fitted with different frames. Where (a) represents the analytical solution field. (b)(d)(e) shows the implementation results of vanilla PIRBN, PINN and FE-PIRBN respectively. (c) and (f) respectively show the absolute error comparison between the original PIRBN and FE-PIRBN frames (the absolute field distribution of the field value subtraction of the same coordinate point).

Table 2

Single cylinder fitting of pinn and Pinn with hash-encoding(named pinn-hash) when wave number k ranges from 2 to 4. The first column is the incident wave frequency, here expressed as wave number k . The second and third columns represent analytical solutions and **feko** numerical simulation solutions, respectively. The fourth and fifth columns represent the output solutions of pinn and pinn-hash. The last column represents the number of input points and the relative error of the neural network. n_x and n_y represent the number of points in the x and y directions, respectively. Error1, error2, and error3 represent the relative errors of **feko**, pinn, and pinn-hash relative to the analytic solution, respectively.

freq	analytic	feko	pinn	pinn-hash	init error
$k=2$					$n_x * n_y = 60*60$ error1 = 0.39% error2 = 1.61% error3 = 1.51%
$k=3$					$n_x * n_y = 60*60$ error1 = 0.41% error2 = 23.9% error3 = 3.88%
$k=4$					$n_x * n_y = 60*60$ error1 = 0.42% error2 = 23.79% error3 = 17.86%

algorithm and the analytic solution. **feko** employs 8 threads for parallel computing. And then the complex scattering cases of double cylinders are tested for electromagnetic trend testing, i.e., whether the physically driven model can roughly fit the complex model, which is a multi-target body electromagnetic field fitting that has not been attempted by the once physically driven model. Due to the difficulty of solving the analytical solution for the multi-objective task, the error comparison of the double-cylinder scattering problem only uses the **feko** numerical solution as a reference.

The quantity of input coordinate points is a critical factor influencing the convergence of neural network outcomes. Based on empirical observations, this data is designated as n_{wave} when 7 to 30 points are selected in each direction within a wave period. The number of data points in the same direction is defined as $int[length * n_{wave} / \lambda]$, where $length$ denotes the extent of the region and λ represents the wavelength of the incident electromagnetic wave, expressed as $2\pi/k$. Consequently, the final expression for one-dimensional point selection becomes $int[length / 2\pi * (n_{wave} * k)]$. For example, if $k = 2$, with a region length

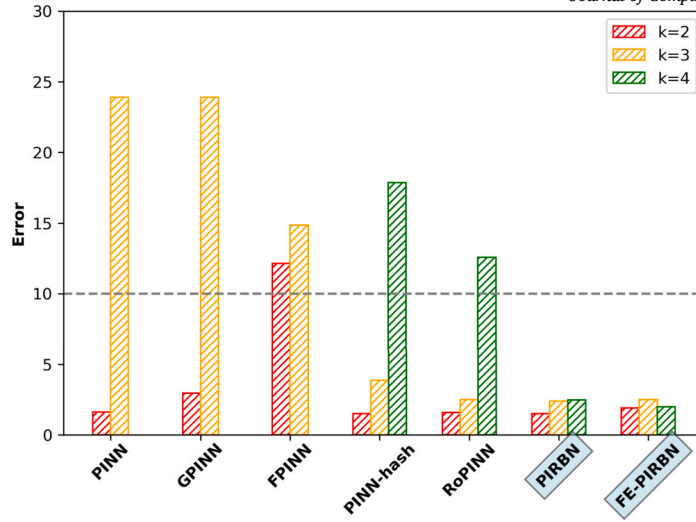


Fig. 6. Comparison of scattering results of seven kinds of physical information neural network models for ideal conductor cylinders under three different frequency incident waves. The ordinate shows the error between the fitting result and the analytical solution. The horizontal coordinate is the result of different models. The meaning of the gray line segment is that when the error exceeds 10, the scattering results of the current frequency cannot be correctly fitted in the model, so the calculation of the next frequency will not be carried out. All models perform well at $k=2$, while the first three models have large errors at $k=3$. Pinn-hash and RoPINN have large errors when $k=4$. Both radial-basis networks can fit the three lower frequency scattering problems shown.

of 2π and $n_{wave} = 30$, then the total number of points in each direction amounts to $k * n_{wave} = 2 * 30$. Thus, in the first row of Table 2, take the point $n_x * n_y = 60 * 60$.

The setup of exogenous fields for electromagnetic field problems is more complicated. It should be emphasized that the boundary conditions used to truncate the computational region should only absorb the scattered field and not the total field. Therefore, we use the total field decomposition method, by which the total field is separated into scattered and incident components. Subsequently, we formulate Maxwell's equations for the scattered field, ultimately deriving the partial differential equations for the region. Here the quantity to be solved for our Helmholtz equations is modified to the scattered field $u = u_0 + iu_1$. Define an exogenous incident plane wave on the surface of an ideal conductor. For a plane wave with a defined direction of incidence, define its beam $\vec{k} = k_0$. The final Helmholtz equation to be solved is:

$$-u_{xx} - u_{yy} - k_0^2 u = 0, \quad \Omega = \overline{B(0, R)} \quad (14)$$

Define Newman's Type II boundary conditions on the surface of an ideal conductor:

$$\gamma_1 u := \nabla(u \cdot n) = -u^{inc}, \quad (x, y) \in \Gamma^{in} := \partial(B(0, R)) \quad (15)$$

where n denotes the normal vector of the target body. In the standard case of **deepxde**, the outer region boundary conditions are applied as first order absorbing boundary conditions. Here we optimize, as mentioned earlier, in the outer region of the scattering problem we apply a perfect matching layer as a perfect way to fit the open domain information.

3.2.1. Cylindrical scattering problem of single ideal conductor

Generally speaking, PINN-related literature is optimized from three aspects. The first is the design of the loss term [61]. The typical method is to add the higher-order constraint residual of PDE to the loss function. Since the residuals usually fluctuate around zero, penalizing the slope of the residuals can reduce the volatility and bring the residuals closer to zero. The second is the transformation of the input feature, mainly through the construction of a PDE solution, the input of the network into the shape of similar features. For example, if the solution $u_i(x)$ of PDE is a periodic function of period P with respect to x_j , then $u_i(x)$ can be decomposed into a weighted summation of the Fourier basis function [52] in the direction of x_j , and the characteristic transformation of the input can better capture the properties of the periodic solution. Furthermore, if PDE is a complex problem, it is a good choice to convert the traditional single coordinate points to high-dimensional eigenvectors [62]. Finally, the training method optimization. The traditional PINN loss is defined only at a series of selected points [16], while the training paradigm of region optimization allows the optimization process to be extended from the selected scatter points to their neighborhoods, theoretically reducing the generalization error of the entire domain [63]. Therefore, we choose four representative methods of the above three methods, GPINN [61], FPINN [52], PINN-hash [62] and RoPINN [63], as our reference models to observe their performance under the high-frequency electromagnetic scattering problem.

Table 2 shows the fitting results of pinn and pinn-Hash methods for wave number k from 2 to 4, that is, frequencies from 95.49MHZ to 190.99MHZ, in the single cylindrical conductor problem. The first column is the analytic solution, the second is the **feko** numerical simulation results, the third and fourth columns are the output solutions of pinn and pinn with hash-encoding, and the fifth column

shows the input collocation points of each neural network method and the relative errors of **feko** and neural network compared with the analytic solution.

$$error = \frac{\sum |y_{out} - y_{ana}|}{\sum y_{ana}} * 100\% \quad (16)$$

Eq. (16) shows the calculation method of relative error. Where y_{out} represents the set of numerical solutions calculated by **feko** using MOM, or the set of solutions of the neural network. y_{ana} represents the set of analytic solutions for a single cylinder.

As can be seen from Table 2, when the wave number is 4, that is, the frequency is 190.99MHZ, pinn and pinn-hash networks cannot be accurately fitted. According to the statement [62], pinn-hash speeds up the network convergence process by reducing the depth and width of the fully connected network, but the final accuracy is almost the same as the original pinn. Therefore, pinn-hash is difficult to work on higher frequency problems.

Fig. 6 shows a comparison bar chart of the seven models. All models perform well at $k=2$, while the first three models have large errors at $k=3$. Pinn-hash and RoPINN have large errors when $k=4$. Both radial-basis networks can fit the three lower frequency scattering problems shown. As the electromagnetic problem is a complex problem, under normal circumstances, the losses include six losses, including initial conditions, boundary conditions and PDE. If the higher-order constrained residual of PDE is added to the loss term, at least four additional losses will be caused in the two-dimensional case. Therefore, PINN network is more likely to fall into local optimal when facing optimization objectives with up to ten losses. This is one of the main reasons why GPINN fails. The transformation of input features and the optimization of training methods can only optimize the ability of the network to capture coordinate features, but the disadvantages of the fully connected layer have already emerged. Through Fourier analysis, the MLP layer can inhibit the capture of specific frequency problems [64], so the conversion of simple input features has no obvious effect on high-frequency problems.

Table 3 shows the frequency-increasing generalization ability of the two RBF networks in the case of electromagnetic scattering from a single cylindrical target. Table 3 shows the fitting results of PIRBN and FE-PIRBN methods when k ranges from 2 to 15, that is, when the frequency ranges from 95.49MHZ to 716.2MHZ. The third and fourth columns show the output solutions of PIRBN and FE-PIRBN. It can be seen that the error of the numerical solution will increase with the increase of the incident wave frequency. It can be observed that when the error approaches 5%, the scattered field can no longer be accurately fitted. Hence, it is more rational to control the overall error below 5%. With the increase of frequency, PIRBN's fitting performance is worse, and the error reaches 12.54% at $k=15$. In 14 cases with different frequencies, the maximum error of PIRBN was 12.54%, and the maximum error of FE-PIRBN was 4.61%. In the first 7 cases with small wave number, the average relative error of PIRBN and FE-PIRBN is 4.31% and 2.09% respectively, and the average relative error of FE-PIRBN is about 51.5% lower than that of PIRBN. In the last 7 cases with large wave number, the average relative error of PIRBN and FE-PIRBN is 8.36% and 2.75%, respectively, and the average relative error of FE-PIRBN is about 67.1% lower than that of PIRBN. It is clear that FE-PIRBN achieves a significant increase in accuracy and more robust convergence characteristics in high frequency electromagnetic scattering tasks with cell columns compared to the original PIRBN method. In the case of increasing frequency, FE-PIRBN shows excellent generalization ability.

In order to verify the anti-noise capability of FE-PIRBN model, random Gaussian white noise is added to the inner boundary condition, which is set as follows:

$$\gamma_1 u = -u^{inc} + \varepsilon, \varepsilon \sim N(0, 0.1^2) \quad (17)$$

Where ε represents normally distributed noise with a standard deviation of 0.1.

Fig. 7 shows the noise resistance of FE-PIRBN in the frequency generalization test due to slight bias of the standard incident plane wave. When $K=13$, the maximum error is 7.21%, when $K=2$, the minimum error is 2.54%, and the overall average error is 4.98%. In particular, with the increase of frequency, the noise slightly affects the fitting effect of the network, but the overall error with the **feko** numerical solution is still less than 7%. In the case of the single metal cylinder, the model can show a good electromagnetic scattering pattern if the total field solution fitting error does not exceed 10%, so the noise resistance is verified. The reason may be that the numerical deviation of the standard incident wave on the scattering boundary caused by noise leads to the poor fitting result of the whole electromagnetic field.

3.2.2. Cylindrical scattering problem of double ideal conductors

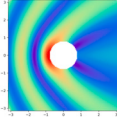
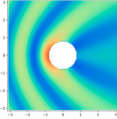
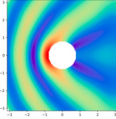
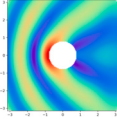
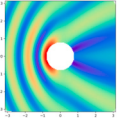
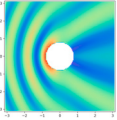
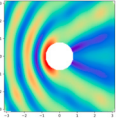
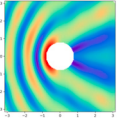
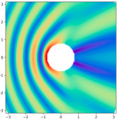
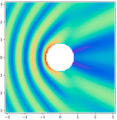
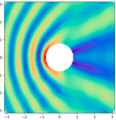
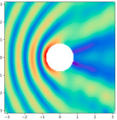
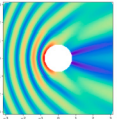
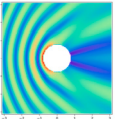
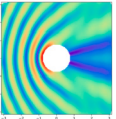
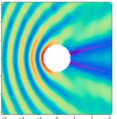
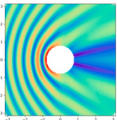
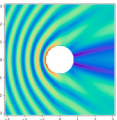
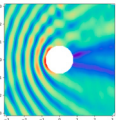
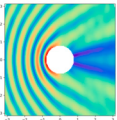
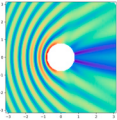
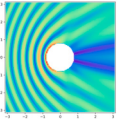
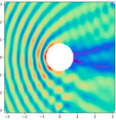
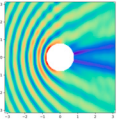
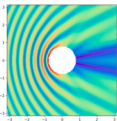
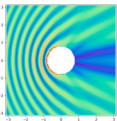
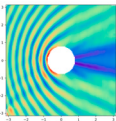
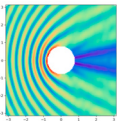
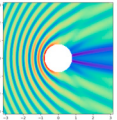
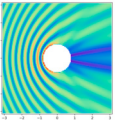
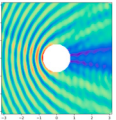
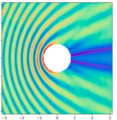
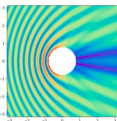
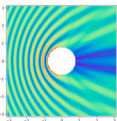
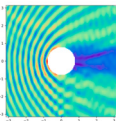
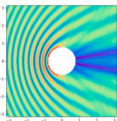
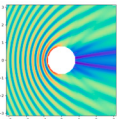
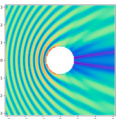
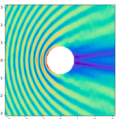
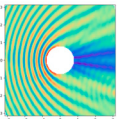
Table 4 shows the fitting results of pinn and pinn-hash methods for the two-cylinder conductor problem when k is from 2 to 4. When the wave number k is 4, that is, the frequency is 190.99MHZ, pinn and pinn-hash cannot be accurately fitted.

Table 5 shows the increased frequency generalization ability of the two RBF networks in the case of electromagnetic scattering of a double cylindrical target. Table 5 shows the scattering field fitting results of PIRBN and FE-PIRBN methods when k ranges from 2 to 8, that is, the frequency ranges from 95.49MHZ to 381.97MHZ. Columns three and four are the output solutions of PIRBN and FE-PIRBN. Since the method of moments ultimately involves solving for a dense matrix, the triangular mesh will significantly expand as the number of targets increases or the shapes of targets become more complex. When k is greater than 8, the current computing resources are unable to utilize more memory to continue computing the scattering field. Therefore, the comparison test had to be interrupted at $k=8$, i.e., $f=381.97$ MHZ. However, the calculation of neural network is only related to the display memory size of the input coordinate point when the operation is carried out, and it can still be calculated when the wave number is larger.

Table 5 shows that the overall error between PIRBN and FE-PIRBN in complex cases is larger than that in single-cylinder cases. As the frequency rises, the fit of PIRBN to the rear side field of the cylinder becomes increasingly indistinct. In the seven cases

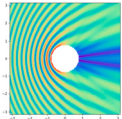
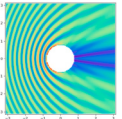
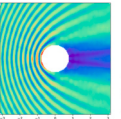
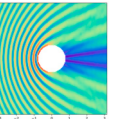
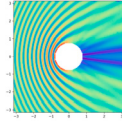
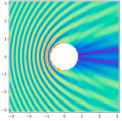
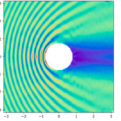
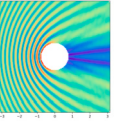
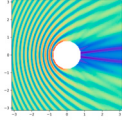
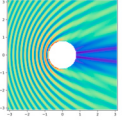
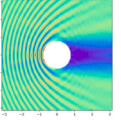
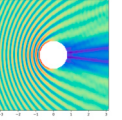
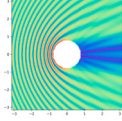
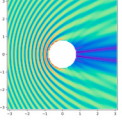
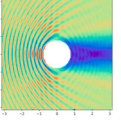
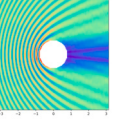
Table 3

The electromagnetic scattering generalization test of FE-PIRBN under a single cylindrical target. The fitting of PIRBN and FE-PIRBN single cylinder scattering problem when the wave number k ranges from 2 to 15.

freq	analytic	feko	PIRBN	FE-PIRBN	init error
k = 2					$nx * ny = 60 * 60$ error1 = 0.39% error2 = 1.51% error3 = 1.92%
k = 3					$nx * ny = 60 * 60$ error1 = 0.41 error2 = 2.39% error3 = 2.49%
k = 4					$nx * ny = 60 * 60$ error1 = 0.48% error2 = 2.47% error3 = 2.0%
k = 5					$nx * ny = 65 * 65$ error1 = 0.50% error2 = 5.2% error3 = 1.99%
k = 6					$nx * ny = 72 * 72$ error1 = 0.70% error2 = 7.01% error3 = 1.59%
k = 7					$nx * ny = 77 * 77$ error1 = 0.74% error2 = 6.29% error3 = 3.03%
k = 8					$nx * ny = 80 * 80$ error1 = 0.73% error2 = 5.35% error3 = 1.62%
k = 9					$nx * ny = 81 * 81$ error1 = 0.74% error2 = 5.57% error3 = 2.99%
k = 10					$nx * ny = 80 * 80$ error1 = 0.77% error2 = 7.47% error3 = 1.40%
k = 11					$nx * ny = 88 * 88$ error1 = 0.89% error2 = 6.42% error3 = 3.62%

(continued on next page)

Table 3 (continued)

freq	analytic	feko	PIRBN	FE-PIRBN	init error
k = 12					$nx * ny = 84 * 84$ error1 = 0.93% error2 = 7.29% error3 = 1.78%
k = 13					$nx * ny = 91 * 91$ error1 = 0.99% error2 = 8.84% error3 = 1.97%
k = 14					$nx * ny = 98 * 98$ error1 = 1.15% error2 = 10.4% error3 = 2.85%
k = 15					$nx * ny = 105 * 105$ error1 = 1.20% error2 = 12.54% error3 = 4.61%

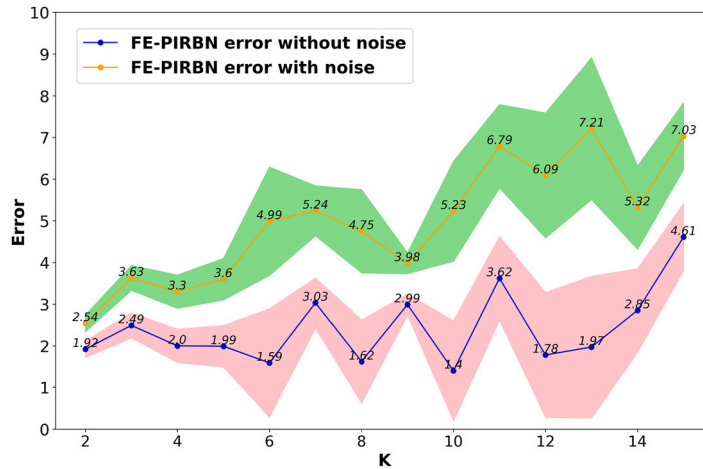


Fig. 7. The anti-noise performance test of the FE-PIRBN model. The orange line represents the percentage error between the model and the numerical solution in the presence of noise, while the blue line indicates the percentage error in the absence of noise. Specifically, the average error with noise is 4.98%, whereas the average error without noise is 2.41%. The green and pink ribbon color bars represent the error difference between the noisy model and the noiseless model at the same wave number (frequency). The narrower the color bar, the smaller the error difference.

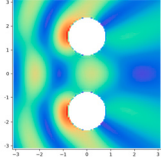
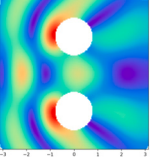
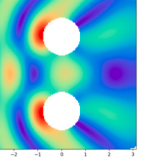
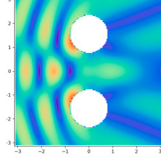
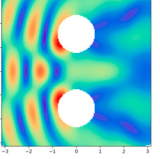
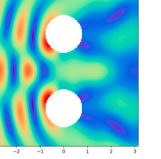
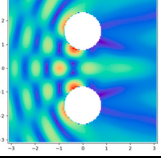
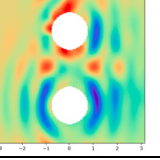
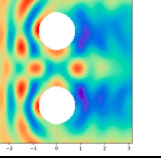
illustrated, the maximum PIRBN fitting error is 17.29%, while the maximum FE-PIRBN fitting error is 5.82%. The average error of PIRBN is 8.92%, the average error of FE-PIRBN is 3.95%, and the average improvement of FE-PIRBN is about 55.72%.

3.2.3. Performance analysis

FE-PIRBN has a better ability to capture high-frequency scattering fields, and can be further optimized in training efficiency compared to vanilla PIRBN. Table 6 shows the performance comparison of **feko** numerical simulation and neural network proxy model for solving double-cylinder scattering problems. It can be observed that as the frequency increases, the number of discrete grids in the numerical simulation doubles, thereby doubling the memory consumption and prolonging the calculation time. The major time consumption is manifested in the calculation of near field. When k is greater than 8, the current cpu environment does not support further computing. The forward solution of the neural network model is not affected by the number of grids, as long as the number of input coordinate points is appropriate, which has been discussed in Section 3.1.1, the scattering field of higher frequencies can be calculated. Due to the more complex network structure of FE-PIRBN, the number of parameters during the training process will be greater. For the scattering problem at the same frequency, the solution time of FE-PIRBN does not possess a significant

Table 4

Pinn and pinn-hash double cylindrical scattering fit for wavenumber k from 2 to 4. The first column is the incident wave frequency, here expressed as wave number k . The second column **feko** numerical simulation solution. The third and fourth columns represent the output solutions of pinn and pinn-hash. The last column represents the number of input points and the relative error of the neural network. nx and ny represent the number of points in the x and y directions, respectively. error1 and error2 represent the relative errors of pinn and pinn-hash relative to the numerical solution, respectively.

freq	feko	pinn	pinn-hash	init error
$k=2$				$nx * ny = 60 * 60$ error1 = 2.92% error2 = 2.41%
$k=3$				$nx * ny = 60 * 60$ error1 = 5.70% error2 = 7.97%
$k=4$				$nx * ny = 60 * 60$ error1 = 28.92% error2 = 29.89%

advantage. For example, as shown in Table 5, when $k = 3$, the output relative errors of PIRBN and FE-PIRBN are similar, 5.79% and 4.18%, respectively, but PIRBN is 35.16% faster than FE-PIRBN shown in Table 6. When the frequency is gradually increased, the shortcomings in time will be masked. For example, when $k=6$, the accuracy of FE-PIRBN is about 55.36% higher than that of PIRBN, but the time is only 22.3% slower. Therefore, FE-PIRBN has more advantages than PIRBN if it is needed to accurately fit the high-frequency scattering field.

Fig. 8 shows the line of error between the single/double cylinder scattering field predicted by four neural network models and the feko numerical simulation results. The blue and yellow lines indicate the fitting error of the Pinn-class model. When $k = 4$, both PINN models in both scattering cases diverge, with an error as high as 23.79% in the single-cylinder scattering case shown in Fig. 8 (a) and 29.89% in the double-cylinder scattering case shown in Fig. 8 (b). As the frequency continues to increase, it makes no sense to continue to predict with the PINN model, so a vertical dotted line separates the prediction error representation of the PINN model from that of the PIRBN model. The green and red lines indicate the fitting error of PIRBN and our FE-PIRBN model. In both scattering cases, the errors will gradually increase with the increase of frequency, and the more complex the model, the higher the errors. The green padding indicates the decrease in error of FE-PIRBN compared to PIRBN. The higher the frequency, the more obvious the improvement. In the error chart for the single-cylinder case presented in (a), the error between FE-PIRBN and PIRBN is most pronounced at $k = 15$, with values of 4.61% and 12.54%, respectively, resulting in an improvement magnitude of 63.24%. In the error diagram for the double-cylinder case illustrated in (b), the error between these two models reaches its peak at $k=7$, showing values of 3.12% and 15.51%, respectively, leading to an enhancement amplitude of 79.88%.

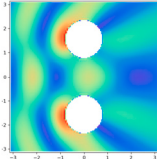
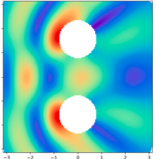
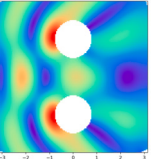
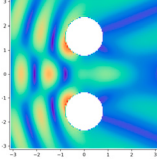
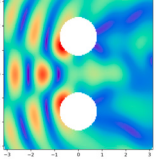
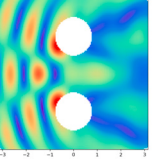
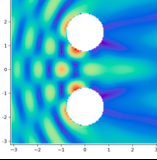
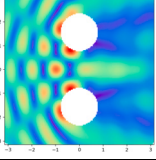
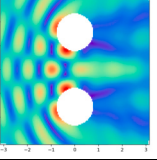
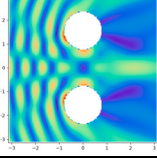
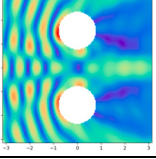
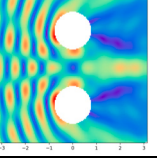
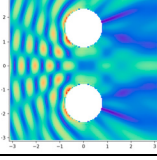
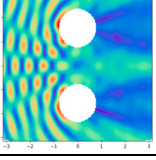
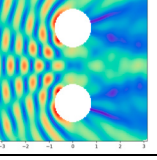
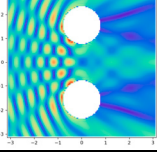
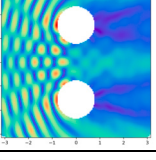
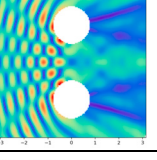
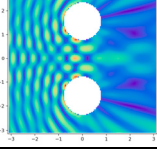
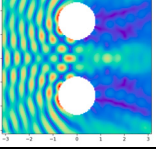
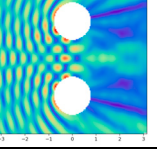
4. Conclusion

This study introduces the Feature-Enhanced Physics-Informed Radial Basis Network (FE-PIRBN) model, designed to capture high-frequency electromagnetic characteristics with greater precision and fewer inputs. We have examined the limitation of the vanilla PIRBN in addressing high-frequency electromagnetic challenges and demonstrated that the integration of a feature enhancement module with hash encoding significantly enhances the network's ability to predict high-frequency features.

The FE-PIRBN model was applied across a range of test cases, including the fitting of two dimensional incident wave field and scattering scenarios of single and double cylinder ideal conductors. In the single cylinder problem, FE-PIRBN achieved a maximum error of 4.61%, with an average relative error of 2.09% in the first seven cases featuring smaller wave numbers, which is about 51.5% lower than that of the PIRBN. In the latter seven cases with larger wave numbers, FE-PIRBN's average relative error was 2.75%, which is about 67.1% lower than the PIRBN. In the double cylinder problem, the maximum fitting error of FE-PIRBN was 5.82%, with an average error of 3.95%, which is about 55.72% lower than that of PIRBN. Furthermore, our analysis indicates that FE-PIRBN is, on average, 43.44% faster than **feko** numerical simulations. In the case of increasing frequency, FE-PIRBN shows excellent generalization ability. Our experiments also demonstrate that the computational load for high-frequency electromagnetic scattering problems increases exponentially with frequency using numerical simulation methods, whereas the neural network approach remains

Table 5

The electromagnetic scattering generalization test of FE-PIRBN under a double-cylindrical target. The scattering field fitting between PIRBN and FE-PIRBN with wave-number k from 2 to 8.

freq	feko	PIRBN	FE-PIRBN	init error
$k=2$				$nx * ny = 60 * 60$ error1 = 3.58% error2 = 3.25%
$k=3$				$nx * ny = 60 * 60$ error1 = 5.79% error2 = 4.18%
$k=4$				$nx * ny = 60 * 60$ error1 = 5.34% error2 = 3.66%
$k=5$				$nx * ny = 65 * 65$ error1 = 5.88% error2 = 3.55%
$k=6$				$nx * ny = 72 * 72$ error1 = 9.05% error2 = 4.04%
$k=7$				$nx * ny = 77 * 77$ error1 = 15.51% error2 = 3.12%
$k=8$				$nx * ny = 80 * 80$ error1 = 17.29% error2 = 5.82%

relatively stable. This observation underscores the significant advantage of the FE-PIRBN model in high-frequency electromagnetic scattering computations.

Our experiments have preliminarily confirmed the significant advantages of combining hash-encoding with PIRBN in high-frequency electromagnetic field prediction. However, the practical application of this method in electromagnetic engineering calculations still has many directions that need to be further explored. For instance, efficient calculation of radar cross-section (RCS) under complex geometric models, calculations of electromagnetic scattering for aircraft, and other typical problems of electrically large scale. Additionally, enhancing the architecture and training methods to improve the generalization capability of physics-informed neural networks, especially in predicting accuracy under significant geometric model variations, presents an interdisciplinary challenge at the intersection of machine learning and electromagnetic computation.

Table 6

Performance comparison of **feko** numerical simulation and neural network model for solving double-cylinder scattering problems. The first column is the wave number k , and the second, fifth and sixth columns are the solution times (in s) for **feko**, PIRBN and FE-PIRBN, respectively. The third and fourth columns represent the number of discrete grids and memory consumption (in M/G Byte) of **feko** at the current frequency, respectively. The parentheses in the fifth and sixth columns indicate the percentage improvement of the training time of the neural network compared to the numerical simulation time at the same frequency.

k	feko			PIRBN time(s)	FE-PIRBN time(s)
	time(s)	mesh	memory (Byte)		
2	189.19	1288	99.459(M)	258.39 (-36.58%)	401.80 (-112%)
3	429.70	2904	342.451(M)	258.59 (39.82%)	398.80 (7.19%)
4	1209.90	5328	924.469(M)	260.24 (78.49%)	437.72 (62.82%)
5	2070.76	8136	1.923(G)	346.58 (83.43%)	479.15 (76.84%)
6	4124.68	11948	3.773(G)	466.54 (88.69%)	600.18 (85.45%)
7	7736.15	16288	6.623(G)	582.03 (92.48%)	720.41 (90.69%)
8	11258.75	21284	14.391(G)	658.60 (94.15%)	779.61 (93.08%)

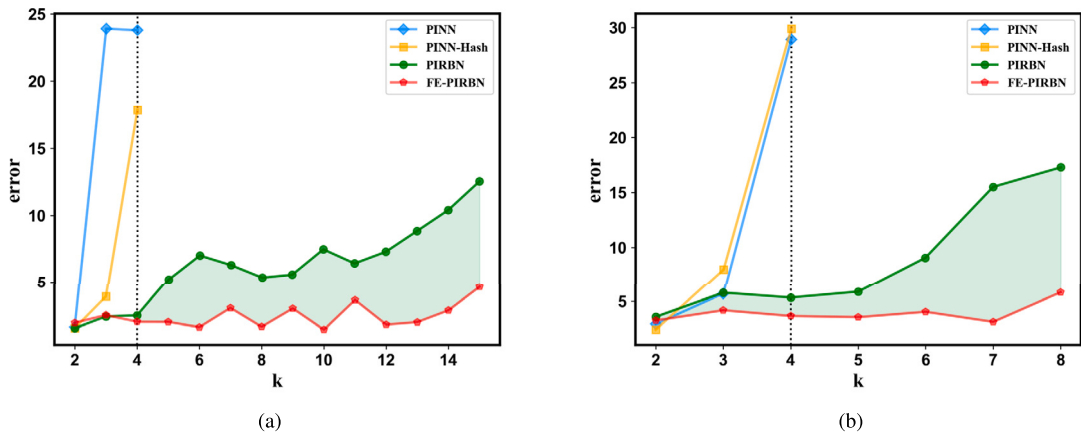


Fig. 8. The scattering fitting errors of four deep learning models for single/double cylinder targets at different frequencies. The horizontal coordinate represents the frequency, denoted by wave number k . The ordinate represents the percentage error. (a) and (b) show the error line graphs of the scattering field of single and double cylindrical targets respectively. The horizontal coordinate represents the frequency, the wave number k , and the vertical coordinate represents the percentage of error. Where the vertical dashed line at $k=4$ indicates the frequency at which the Pinn-class model begins to diverge. The light green filling coating indicates the error difference between PIRBN and FE-PIRBN.

CRedit authorship contribution statement

Huajian Zhang: Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chao Li:** Supervision, Methodology. **Rui Xia:** Methodology, Investigation, Conceptualization. **Xinhai Chen:** Software, Methodology, Formal analysis. **Tiaojie Xiao:** Formal analysis, Conceptualization. **Xiao-Wei Guo:** Writing – review & editing, Funding acquisition, Data curation, Conceptualization. **Jie Liu:** Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiao-Wei Guo reports was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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